

Vector Differentiation: Gradient of a Function of Radius

1. Gradient of a Radial Scalar Function

Name of Concept

Gradient of a Function of Distance (r), or the Vector Chain Rule for Spherical Symmetry.

Core Mathematical Formula

$$\nabla f(r) = f'(r) \frac{\vec{r}}{r}$$

Beginner-Friendly Explanation of Operations

- The Position Vector ($\vec{r}$):** This represents a vector pointing from the origin $(0, 0, 0)$ to any point space (x, y, z) . It is written as $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.
 - The Magnitude (r):** The scalar letter r (without the bar) represents the straight-line distance from the origin to that point. It is computed using the 3D Pythagorean theorem: $r = \sqrt{x^2 + y^2 + z^2}$.
 - The Single-Variable Derivative ($f'(r)$):** Since $f(r)$ depends *only* on the distance r and not directly on individual x, y, z coordinates, $f'(r)$ is just the standard derivative of that function with respect to r (like differentiating x^2 to get $2x$).
 - The Unit Position Vector ($\frac{\vec{r}}{r}$):** This is a vector of length 1 pointing directly outward from the origin. The formula shows that the gradient of any radial function points straight outward, and its strength depends on the regular derivative $f'(r)$.
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2. Analogous Example (The Chain Rule)

To understand why the derivative splits into two parts, remember the standard calculus Chain Rule.

If you differentiate a nested function like $\sin(x^2 + 1)$ with respect to x :

- First, you differentiate the outer function: $\sin(\dots) \rightarrow \cos(\dots)$
- Second, you multiply by the partial derivative of the inner function: $\frac{\partial}{\partial x}(x^2 + 1)$

$$\frac{\partial}{\partial x} \sin(x^2 + 1) = \cos(x^2 + 1) \cdot 2x$$

We use this exact same logic for $\frac{\partial}{\partial x} f(r)$. Because r depends on x , you take the derivative of the outer function $f(r)$ with respect to r (which is $f'(r)$), then multiply it by the partial derivative of the inside variable with respect to x (which is $\frac{\partial r}{\partial x}$).

3. Step-by-Step Proof

Question

Prove that $\nabla f(r) = f'(r) \frac{\vec{r}}{r}$.

Step 1: Establish the relationship between coordinates and radius

Define the position vector $\vec{r}$ and its scalar magnitude r :

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

Square both sides to eliminate the radical sign:

$$r^2 = x^2 + y^2 + z^2$$

Step 2: Differentiate the radius relationship partially with respect to x

Differentiate both sides with respect to x , treating y and z as constants. Remember to use the chain rule on the left side ($r^2 \rightarrow 2r \cdot \frac{\partial r}{\partial x}$):

$$2r \frac{\partial r}{\partial x} = 2x + 0 + 0$$

$$2r \frac{\partial r}{\partial x} = 2x$$

Divide both sides by $2r$ to isolate the partial derivative:

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

Step 3: Find the corresponding derivatives for y and z

By following the exact same algebraic steps for the other axes, we find symmetric results:

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

Step 4: Apply the vector gradient operator to $f(r)$

The general definition for the gradient of any scalar field is:

$$\nabla \phi = \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z}$$

Substitute $\phi = f(r)$ into the operator equation:

$$\nabla f(r) = \vec{i} \frac{\partial f(r)}{\partial x} + \vec{j} \frac{\partial f(r)}{\partial y} + \vec{k} \frac{\partial f(r)}{\partial z}$$

Step 5: Expand each component using the chain rule

Apply the chain rule to each partial derivative term:

$$\frac{\partial f(r)}{\partial x} = f'(r) \frac{\partial r}{\partial x}$$

$$\frac{\partial f(r)}{\partial y} = f'(r) \frac{\partial r}{\partial y}$$

$$\frac{\partial f(r)}{\partial z} = f'(r) \frac{\partial r}{\partial z}$$

Substitute these back into our full gradient expression:

$$\nabla f(r) = \vec{i} \left[f'(r) \frac{\partial r}{\partial x} \right] + \vec{j} \left[f'(r) \frac{\partial r}{\partial y} \right] + \vec{k} \left[f'(r) \frac{\partial r}{\partial z} \right]$$

Step 6: Substitute the calculated components and simplify

Substitute the specific values for $\frac{\partial r}{\partial x}$, $\frac{\partial r}{\partial y}$, and $\frac{\partial r}{\partial z}$ found in Step 2 and Step 3:

$$\nabla f(r) = \bar{i} \left[f'(r) \frac{x}{r} \right] + \bar{j} \left[f'(r) \frac{y}{r} \right] + \bar{k} \left[f'(r) \frac{z}{r} \right]$$

Factor out the common scalar term $f'(r)$ and the common denominator r :

$$\nabla f(r) = \frac{f'(r)}{r} (x\bar{i} + y\bar{j} + z\bar{k})$$

Step 7: Substitute the original position vector back in

Since $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$, replace the parenthesis term with $\bar{r}$:

$$\nabla f(r) = \frac{f'(r)}{r} \bar{r}$$

Rearrange the terms to match the target layout:

$$\nabla f(r) = f'(r) \frac{\bar{r}}{r}$$